Fore-and-Back for Changepoint Analysis

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Abstract. The abstract should briefly summarize the contents of the paper in 150–250 words.

Keywords: Changepoint analysis · matheuristics · Fore-and-Back.

1 Introduction

Please note that the first paragraph of a section or subsection is not indented. The first paragraph that follows a table, figure, equation etc. does not need an indent, either.

Subsequent paragraphs, however, are indented.

Sample Heading (Third Level) Only two levels of headings should be numbered. Lower level headings remain unnumbered; they are formatted as run-in headings.

Sample Heading (Fourth Level) The contribution should contain no more than four levels of headings. Table 1 gives a summary of all heading levels.

Table 1. Table captions should be placed above the tables.

Heading level	Example	Font size and style
Title (centered)	**Lecture Notes**	14 point, bold
1st-level heading	**1 Introduction**	12 point, bold
2nd-level heading	**2.1 Printing Area**	10 point, bold
3rd-level heading	**Run-in Heading in Bold.** Text follows	10 point, bold
4th-level heading	*Lowest Level Heading.* Text follows	10 point, italic

Please try to avoid rasterized images for line-art diagrams and schemas. Whenever possible, use vector graphics instead (see Fig. 1).

For citations of references, we prefer the use of square brackets and consecutive numbers. Citations using labels or the author/year convention are also acceptable.

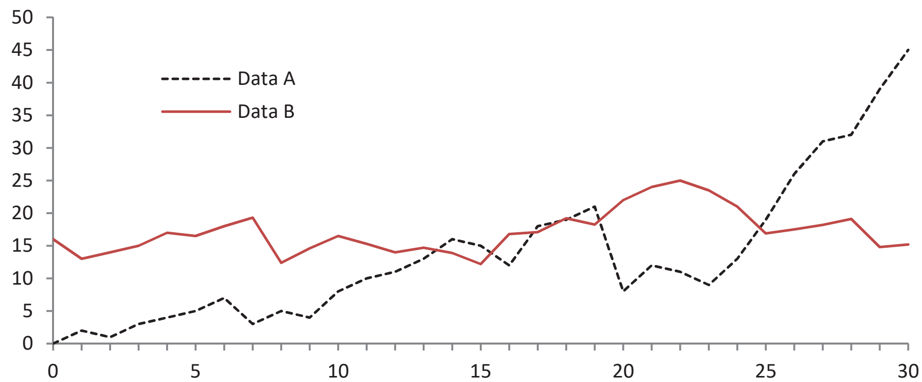

Fig. 1. A figure caption is always placed below the illustration. Please note that short captions are centered, while long ones are justified by the macro package automatically.

2 Changepoint analysis

Letteratura etc. [3]

3 Fore-and-Back for changepoint analysis

Fore-and-back [1, 3] is an extension of *Beam Search* (BS) that can improve its effectiveness. Fore-and-Back, when run with no limits on computational resources, becomes an exact solution method. However, by design, it is mainly concerned with heuristic solving, trying to quickly get high quality solutions with little attention paid to optimality proofs.

A significant characteristic of this method is that, despite being a primal only method, it is able to compute bounds to the cost of completing partial solutions, therefore to discard partial solutions from expansion and ultimately to reduce the search space.

Beam Search [2]

4 Computational results

Computational results

5 Conclusions

The conlusions

Algorithm 1: Algorithm Fore-and-Back

```
function Fore-and-Back($\delta$, $maxn$, $maxtnodes$);
Input  : $\delta$, beam width, $maxn$, max total num of nodes, $maxtnodes$, max
         num of nodes per tree
Output: A feasible solution $\mathbf{x}^*$ of value $z_{best}$
initialize $t = 0$, $noimpr = 0$, $nNodes = 0$, $z_{best} = \infty$;
while $nNodes \leq maxn$ do        // Alternate forward and backward trees
    $t = t + 1$; $noimpr = noimpr + 1$;
    let $L_0^t = L_{n+1}^t = \{\emptyset\}$, $ntNodes$=0;
    foreach level $h = 1, ..., n$ do
        set $T_h = \{\emptyset\}$;                    // generate the node set $T_h$
        if $t$ is odd then
            set $k = h$; $k1 = h - 1$;
        else
            set $k = n - h + 1$; $k1 = n - h + 2$;
        end
        foreach node $X \in L_{k-1}^t$ do
            foreach component $s \in S_k$ do
                let $X' = X \cup \{s\}$;
                if $X'$ meets conditions $i)$ and $ii)$ then
                    set $T_k = T_k \cup X'$;
                    $ntNodes = ntNodes + 1$;
                    $nNodes = nNodes + 1$;
                    if $h = n$ and $c(X') < z_{best}$ then    // feasible solution
                        set $z_{best} = c(X')$, $noimpr = 0$ and $\mathbf{x}^* = X'$;
                    end
                end
            end
            set $E^k = E^k \cup X$;                        // node expanded
        end
        foreach node $X \in T_k$ do          // extract the subset $L_k^t \in T_k$
            if $|T_k| \leq \delta$ then
                set $L_k^t = T_k$;
            else
                let $L_k^t$ contain only the $\delta$ least cost partial solutions of $T_k$;
            end
        end
        if $ntNodes > maxtnodes$ then break;
    end
    if $noimpr = 2$ then stop ; // sol. not improved for two iterations
end
```

References

1. Bartolini, E., Maniezzo, V., Mingozzi, A.: An adaptive memory-based approach based on partial enumeration. In: Maniezzo, V., Battiti, R., Watson, J.P. (eds.) LION 2, LNCS 5313, pp. 12–24. Springer (2008)
2. Blum, C.: Beam-ACO for simple assembly line balancing. INFORMS Journal on Computing **20**(4), 618–627 (2008)
3. Maniezzo, V., Boschetti, M.A., Stützle, T.: Matheuristics: Algorithms and Implementations. EURO Advanced Tutorials on Operations Research, Springer (2021). https://doi.org/10.1007/978-3-030-64219-4